Table 1. Codebook for Web Search Learning: Challenges and Frustrations

Code	Theme Name	Explanations
NC-N1	Information overload / hard to filter	overwhelming; too much information; flood of results; large volume; excessive scrolling
NC-N2	Poor targeting or lack of specificity	not specific; vague; generic; irrelevant results
NC-N3	Time-consuming search process	time-consuming; many websites; hard to locate answers; paywalls
NC-N4	Credibility or freshness concerns	outdated information; reliability concerns; need to verify; trust issues
NC-N5	Requires substantial prior knowledge	assumes background knowledge; prerequisites; unclear starting point
NC-N6	Lack of code support or interactivity	no code examples; cannot input code; no interactive debugging
NC-N7	Lack of feedback or confirmation	no indication whether I am on the right track
NC-N8	Shallow coverage or limited learning depth	only answers what is asked; no background; no further resources or related ideas
NC-N9	Query formulation burden	results depend heavily on phrasing; requires repeated rewording
NC-N10	Redundant or homogeneous answers	many sources provide similar explanations; little variation
NC-N11	Difficulty applying information to examples	unclear how to apply explanations to the given code
NC-N12	Lack of personalization	information not tailored to my experience or learning needs

Table 2. Codebook for Web Search Learning: Benefits and Strengths

Code	Theme Name	Explanations
NC-P1	Rich and in-depth information	detailed articles; in-depth explanations; diagrams; videos
NC-P2	Multiple perspectives and community vetting	multiple viewpoints; forum discussions; upvotes; community agreement
NC-P3	Improved learning or retention	learn more; better retention; forced to read and process information
NC-P4	Trustworthy and traceable sources	clear citations; identifiable sources; verifiable references
NC-P5	Promotes active thinking and research skills	engagement with the research process; learning by doing
NC-P6	Familiar and easy-to-use interfaces	easy to query; familiar tools; less concern about phrasing
NC-P7	Efficient for known lookups	quick access when the relevant resource is already known
NC-P8	Direct answers from authoritative sources	straightforward responses to specific questions
NC-P9	Easier access to concrete examples	examples readily available; sometimes better than AI-generated ones
NC-P10	Sense of community and collaboration	learning with others; discussion; asking for help

Table 3. Codebook for LLM Learning: Challenges and Frustrations

Code	Theme Name	Explanations
LLM-N1	Generic or shallow responses	general; vague; superficial explanations
LLM-N2	Lack of concrete examples or context	no examples; insufficient contextual grounding
LLM-N3	Accuracy or hallucination concerns	hallucination; uncertainty; need to verify correctness
LLM-N4	Excessive information causing confusion	long lists; too many issues at once; overwhelming

Code	Theme Name	Explanations
LLM-N5	Off-topic or drifting responses	irrelevant content; loss of context
LLM-N6	Need for repeated follow-up queries	iterative questioning; clarification loops
LLM-N7	Prompting skill requirements	effective prompting needed; reliance on prior knowledge
LLM-N8	Response latency	waiting time for responses
LLM-N9	Cognitive load to interpret responses	takes time to digest or understand explanations
LLM-N10	Overly wordy or unnatural tone	verbose; formal; unnatural language
LLM-N11	Insufficient personalization	does not adapt to my experience or expertise
LLM-N12	Single-source limitation	cannot easily obtain multiple viewpoints
LLM-N13	Missing code navigation support	no line numbers; difficult to reference code locations
LLM-N14	Reduced critical engagement	discourages independent reasoning
LLM-N15	Patronizing or condescending tone	talks down to the user; feels infantilizing
LLM-N16	Narrow topical focus	limited exploration beyond the immediate question

Table 4. Codebook for LLM Learning: Benefits and Strengths

Code	Theme Name	Explanations
LLM-P1	Fast and convenient	fast; quick; time-saving
LLM-P2	Code-specific support and examples	tailored to code; line-by-line explanation; code insertion
LLM-P3	Highly interactive learning	dialogue-based; follow-up questions; conversational
LLM-P4	Clear high-level overview	bird's-eye view; structured summary
LLM-P5	Up-to-date information	current; not outdated
LLM-P6	Easy to understand explanations	clear; accessible; beginner-friendly
LLM-P7	Effective problem resolution	directly solved my problem
LLM-P8	Adaptive and personalized assistance	interprets vague queries; adapts responses dynamically
LLM-P9	High relevance to user intent	specific to my question; non-generic examples
LLM-P10	Multiple solution paths	alternative fixes; different approaches
LLM-P11	Knowledge verification support	non-judgmental feedback; confirmation of understanding
LLM-P12	Adapts to different expertise levels	adjusts explanations for novices and advanced users

Table 5. Codebook for Factors Influencing Participants' Preferences

Code	Theme Name	Decision Cues / Typical Indicators
PF-1	Efficiency / Time-saving	Preference driven by faster access to answers, fewer steps, or more efficient use of time.
PF-2	Ease of access / interaction	Method is easier to interact with (e.g., conversational, intuitive), or requires less effort to ask or follow up.
PF-3	Information clarity / explanation quality	Answers are clearer, more structured, easier to understand, or better explained.
PF-4	Trustworthiness and accuracy	Concern or confidence regarding factual correctness, hallucinations, or reliability of information sources.
PF-5	Depth of learning and support for understanding	Method supports deeper learning, self-paced understanding, tailored explanations, or follow-up guidance.
PF-6	Independence and control	Preference for solving problems independently, controlling the learning flow, or avoiding overreliance on AI assistance.
PF-7	Cognitive load reduction / overwhelm avoidance	Method is preferred because it reduces distraction, information overload, or cognitive burden.

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Code	Theme Name	Decision Cues / Typical Indicators
PF-8	Familiarity, habit, or enjoyment	Preference driven by long-standing habits, familiar tools, personal learning style, or enjoyment of the process.
PF-9	Specificity and contextual fit	Answers are tailored to the user's specific code, problem, or situation, rather than generic examples.
PF-10	Balanced or neutral evaluation	Participant found both methods useful, or suggested using different methods depending on context.
PF-11	Low prompting or query formulation effort	Preference (or neutrality) attributed to not needing to carefully design prompts or queries.
PF-12	Opportunities for further or extended learning	Method provides additional avenues, resources, or ideas that support continued learning beyond the immediate task.